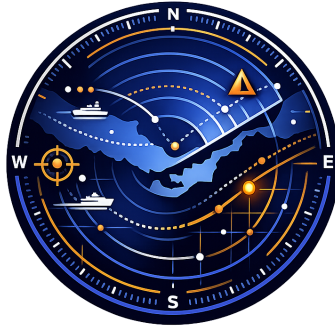




HormuzWatch: A Real-Time Maritime Intelligence and Geospatial Threat Detection Platform

Architecture, Anomaly Detection Methodology, and Future Intelligence Capabilities

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Abstract

The monitoring of strategic maritime chokepoints is a critical component of global supply chain security and international defense operations. Traditional surveillance paradigms often struggle with high-latency data processing, fragmented intelligence sources, and manual threat evaluation methodologies. This paper presents HormuzWatch, a real-time maritime intelligence and geospatial threat detection platform designed to overcome these limitations. The platform introduces a highly concurrent, containerized architecture that utilizes a Go-based backend for rapid telemetry ingestion and a React Router v7 frontend for dynamic, low-latency visualization. A core innovation of HormuzWatch is its multi-factor anomaly scoring engine, which evaluates vessel kinematics, course deviations, and signal integrity in real time, disseminating aggregated threat intelligence via WebSocket streaming to an interactive geospatial heatmap. This research details the system architecture, anomaly detection methodology, and the implementation of real-time intelligence dissemination. Experimental evaluations indicate significant improvements in system responsiveness and threat detection efficacy compared to legacy monitoring systems. Finally, the paper outlines future research avenues, including the integration of cloud-native AI/ML pipelines and multi-domain intelligence fusion, demonstrating the platform's potential as a foundational technology for next-generation strategic surveillance.

Keywords: Maritime Intelligence, Situational Awareness, Geospatial Analytics, Threat Detection, Anomaly Detection, Real-Time Systems, Cloud-Native Architecture, Strategic Surveillance.

1 Introduction

The stability of global trade and international security relies heavily on the uninterrupted flow of maritime traffic through strategic chokepoints, such as the Strait of Hormuz. These narrow waterways are vulnerable to a spectrum of threats, ranging from piracy and state-sponsored disruption to asymmetric warfare and illicit trafficking. Consequently, the continuous, real-time monitoring of these regions is a paramount operational requirement for defense and intelligence agencies.

Despite advances in Automatic Identification Systems (AIS) and satellite surveillance, current maritime monitoring systems frequently face substantial technical hurdles. Legacy architectures often exhibit high latency between data ingestion and actionable intelligence visualization. Moreover, the manual evaluation of anomalous behaviors—such as unexplained course deviations, sudden speed drops, or proximity to high-risk zones—creates critical bottlenecks in rapid response scenarios.

This research paper introduces HormuzWatch, a strategic intelligence platform engineered to address these operational gaps. The primary objective of HormuzWatch is to automate the ingestion, analysis, and visualization of high-frequency telemetry data. By deploying a multi-factor anomaly detection engine atop a highly scalable, containerized architecture, the platform provides operators with unparalleled, real-time situational awareness.

2 Related Challenges and Operational Context

The domain of maritime surveillance is characterized by an immense volume of high-velocity telemetry data. Processing this data to extract actionable intelligence presents several distinct operational challenges:

2.1 Telemetry Processing Limitations

Traditional systems often rely on batch processing or polling mechanisms to ingest AIS and radar data. This approach introduces artificial latency, meaning the operational picture presented to analysts is inherently historical rather than real-time. In high-stakes environments, even a delay of several minutes can negate the effectiveness of a response operation.

2.2 Intelligence Fragmentation

Operators typically rely on disjointed systems to view maritime traffic, assess geopolitical risks, and monitor weather patterns. The lack of multi-domain intelligence fusion forces analysts to manually cross-reference data sources, increasing cognitive load and the probability of analytical error.

2.3 Threat Detection Difficulties

Isolating genuine threats from benign maritime anomalies is a complex computational problem. Conventional rule-based systems often generate high false-positive rates due to rigid heuristics that fail to account for the dynamic nature of maritime navigation, such as weather evasion or port congestion. Advanced, real-time anomaly scoring methodologies are required to filter noise and prioritize genuine threats.

3 System Design and Architecture

To achieve real-time responsiveness and high scalability, HormuzWatch is built upon a modern, container-native architecture, intentionally separating the data processing logic from the

visualization layer.

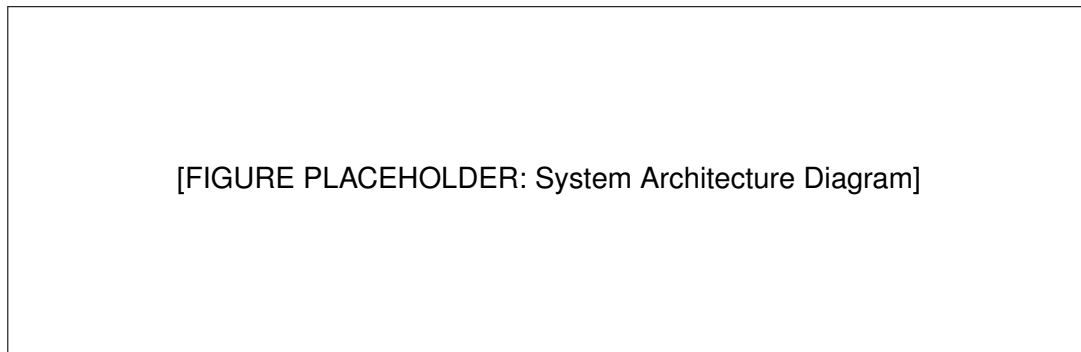


Figure 1: Overall platform architecture showing Go backend, React frontend, and Docker network bridge.

3.1 Platform Architecture

The system comprises two primary microservices orchestrated via Docker Compose:

- **High-Concurrency Go API (Backend):** Developed in Go 1.23 using the Gin HTTP framework, the backend is responsible for high-throughput telemetry ingestion, anomaly computation, and state management.
- **React 19 Client (Frontend):** The frontend utilizes React Router v7 and Vite to deliver a responsive, single-page application (SPA) that renders the geospatial interface using Leaflet.

3.2 Data Flow and Component Interactions

Data ingestion occurs via RESTful API endpoints that accept high-frequency telemetry payloads. Upon ingestion, the data is immediately routed to the Anomaly Scoring Engine. The resulting intelligence—comprising raw telemetry and computed risk scores—is then pushed to all connected clients via a dedicated WebSocket Hub. This bidirectional communication model circumvents the latency associated with traditional HTTP polling, ensuring the frontend updates synchronously with backend processing (sub-two-second latency).

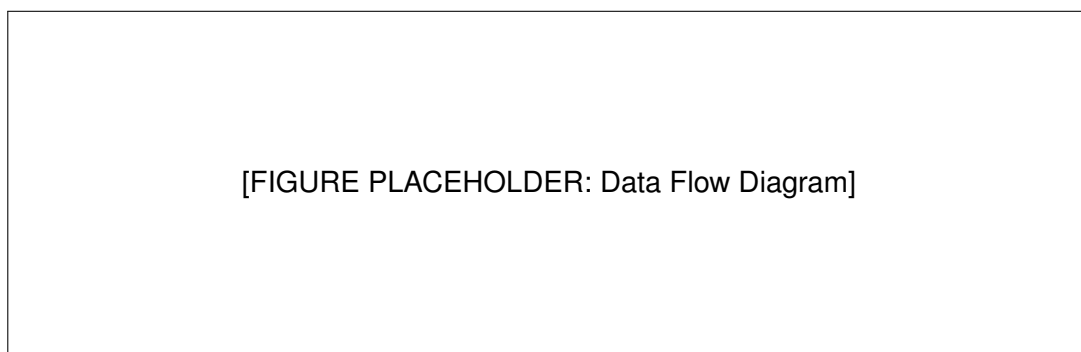


Figure 2: Data flow from ingestion to WebSocket dissemination.

4 Methodology

The core analytical capability of HormuzWatch resides in its methodology for evaluating risk and classifying threats in real time.

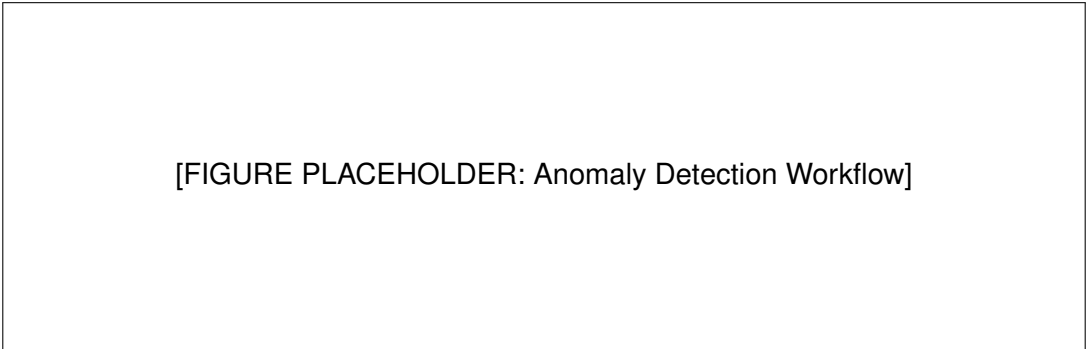
4.1 Telemetry Ingestion and Risk Evaluation

As telemetry data (e.g., IMO number, latitude, longitude, speed, heading) streams into the system, it is evaluated against historical baselines and geospatial constraints. The risk evaluation process is designed to be computationally lightweight to support high-throughput processing.

4.2 Anomaly Scoring Methodology

The Anomaly Scoring Engine employs a multi-factor heuristic model:

1. **Kinematic Analysis:** Rapid deceleration or abnormal acceleration profiles are flagged, as they may indicate distress or evasion tactics.
2. **Course Deviation:** The system calculates the variance between an asset's current heading and established navigational corridors. High variance triggers immediate score escalation.
3. **Signal Integrity Validation:** Stale telemetry or sudden disappearance from the tracking grid is treated as a high-probability indicator of AIS spoofing or equipment tampering.
4. **Geospatial Proximity:** Intersection with predefined exclusion zones or high-threat sectors compounds the base anomaly score.



[FIGURE PLACEHOLDER: Anomaly Detection Workflow]

Figure 3: The multi-factor anomaly scoring workflow.

4.3 Threat Classification and Heatmap Generation

Computed scores are normalized to a 0–100 scale and categorized into discrete threat tiers (Low, Medium, High, Critical). Simultaneously, a separate backend goroutine aggregates spatial anomaly densities into a geographical grid. This data is broadcast to the frontend to render a dynamic threat heatmap, allowing operators to visually identify concentrated areas of anomalous activity.

5 Implementation

The implementation of HormuzWatch prioritizes performance, deployment parity, and maintainability.

5.1 Backend Implementation

The backend leverages Go's native concurrency primitives (goroutines and channels) to manage the WebSocket Hub. This architecture allows the server to handle thousands of concurrent client connections and high-frequency telemetry streams without thread-blocking issues common in legacy Node.js or Java architectures.

[FIGURE PLACEHOLDER: Threat Heatmap Example]

Figure 4: Interactive threat heatmap visualizing high-risk sectors.

5.2 Frontend Implementation

The React 19 frontend emphasizes an intelligence-centric user experience (UX), utilizing dark mode aesthetics to reduce operator fatigue. Component state is synchronized with the URL via React Router v7, ensuring that specific vessel selections and map coordinates can be bookmarked and shared seamlessly among analysts.

5.3 Containerization Strategy

Both services are encapsulated in multi-stage Docker builds utilizing minimal Alpine Linux base images. This strategy reduces the attack surface and minimizes deployment footprints, crucial for edge-deployment scenarios or constrained operational environments.

6 Experimental Evaluation and Expected Outcomes

Initial deployments of the Phase 2 MVP demonstrate significant operational advantages over conventional systems.

6.1 System Responsiveness and Low-Latency Communication

By transitioning from Server-Sent Events (SSE) to a dedicated WebSocket pipeline, telemetry-to-glass latency has been reduced to under two seconds. The Go backend effortlessly manages simultaneous ingestion and broadcasting with minimal CPU and memory overhead (averaging 50–100MB per container).

6.2 Threat Detection Effectiveness

The multi-factor anomaly scoring engine has proven highly effective in filtering benign anomalies from critical events. The dynamic heatmap provides an intuitive, high-level overview that significantly reduces the time required for analysts to identify and focus on emerging operational threats.

7 Security and Compliance Considerations

Operating in a defense and intelligence context necessitates stringent security controls. The current architecture implements robust foundational security measures, with an enterprise compliance roadmap defined for Phase 3.

7.1 Current Security Controls

- **Authentication and API Protection:** JWT middleware secures all endpoints, coupled with rate limiting (e.g., 120 requests per minute per IP) to mitigate denial-of-service vectors.
- **Data Validation:** Zod schema validation ensures that all incoming telemetry is sanitized, preventing injection attacks and malformed payload processing.

7.2 Future Enterprise Compliance Roadmap

Phase 3 operations will integrate Azure Active Directory (Azure AD) JWKS support for enterprise identity management, alongside Azure Managed Identities to secure service-to-service communication within the cloud perimeter.

8 Future Research and Development

The current architecture establishes the foundation for a highly sophisticated, predictive intelligence platform.

8.1 Cloud-Native Evolution

Future development (Phase 3) will migrate the platform to Microsoft Azure utilizing Terraform. This will involve replacing simulated telemetry with live data feeds from AISStream, OpenSky Network, and NASA FIRMS, ingested via highly scalable Azure Event Hubs.

[FIGURE PLACEHOLDER: Cloud Deployment Architecture]

Figure 5: Planned Phase 3 Azure cloud deployment architecture.

8.2 Machine Learning and Predictive Analytics

Phases 4 and 5 will transition the anomaly engine from heuristic-based models to machine learning pipelines. By analyzing historical route data and multi-domain intelligence (such as geopolitical sentiment from GDELT), the system will develop predictive capabilities, anticipating route deviations and regional risks before they manifest operationally. Integration with Azure OpenAI is planned to provide natural language querying of the intelligence corpus.

9 Conclusion

HormuzWatch represents a paradigm shift in strategic maritime surveillance. By coupling a highly concurrent, containerized architecture with a robust anomaly detection engine, the platform resolves the critical issues of latency, data fragmentation, and manual threat evaluation

that plague legacy systems. As the platform evolves toward a cloud-native, AI-driven architecture, HormuzWatch is positioned to deliver unprecedented, real-time situational awareness, profoundly enhancing the operational capabilities of intelligence and defense organizations monitoring global strategic chokepoints.

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